GCSE

Rewarding Learning

Chief Examiner's and Principal Moderator's Report

Digital Technology



Summer Series 2025

Foreword

This booklet outlines the performance of candidates in all aspects of this specification for the Summer 2025 series.

CCEA hopes that the Chief Examiner's and/or Principal Moderator's report(s) will be viewed as a helpful and constructive medium to further support teachers and the learning process.

This booklet forms part of the suite of support materials for the specification. Further materials are available from the specification's microsite on our website at www.ccea.org.uk.

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GCSE DIGITAL TECHNOLOGY

Chief Examiner's Report

Subject Overview

Across all three papers candidates performed well. Responses indicated that they are being given good preparation by their teachers across centres, to enable this. Candidates responded well to the various forms and styles of questions used in the papers. They demonstrated a sound knowledge of the key concepts covered within the specification. All three of the papers allowed for a range of responses and enabled all candidates to display their understanding of the subject content as well as demonstrate their skills in the areas of programming and multimedia.

Following this examination series a number of key points have arisen.

- There is still evidence of candidates not fully reading stimulus material which ultimately impacts negatively on their performance. When answering questions related to stimulus material or scenarios candidates should read this material carefully and reference it, when required, in their answers. This is key because the mark scheme is generally structured to take account of these points. Further detail on this can be found in the unit reports below.
- In open response questions it is important to make use of the correct digital technology terminology. Candidates can miss out on marks by using generic terms such as "faster" or "quicker". This is particularly true in questions where the quality of written communication is assessed.
- It should be noted that in questions where candidates are asked to place a Tick (√) beside a correct statement, in instances where candidates place more than the required number of ticks, only the first n, where n=the required number, will be accepted.
- Where a single response is requested, candidates should not put more than one answer in the space provided. This will result in no marks being allocated.
- Candidates continue to demonstrate solid skills in the area of programming and multimedia which reflects the practical work they undertake as part of these units.

Candidates performed very well in this examination series. They and their teachers are to be commended.

Assessment Unit 1 Compulsory Core

Unit Overview

Overall, the design of the paper enabled candidates to achieve excellent results and accommodated all ability levels. The language used in the paper was appropriate for GCSE, and the mark scheme allowed markers to reward candidates fairly. Feedback from the markers stated that "it was a paper that suited all abilities and covered the full range of topics." Others suggested that "the paper included a high number of questions which were successful in differentiating candidates by ability."

Q1 Multiple Choice

This question consisted of parts a to d, each part contained four short statements, and the candidate was asked to choose the correct response.

(a) Well answered.

(b) Well answered.

(c) Well answered.

(d) Well answered.

Q2 **(a)** This part was generally very well answered by the candidates; the majority got all three correct; however, a few got them in the wrong order and failed to score.

(b) This part again was well answered.

(c) This part was also very well answered.

(d) No issues were identified in answering this part.

Q3 Question three was about data portability and the conversion of data into a specific file format. Question 3 Part (b) was well answered. However, many candidates seemed to struggle with Part (a). This part seemed to challenge most candidates, and it appears that candidates were not prepared in this area of the specification.

Q4 **(a)** Was answered poorly by many. Many candidates failed to identify "the boot up sequence," which takes place when a computer is switched on.

(b) This part was answered well, with only a few candidates mixing the words up.

(c) No real issues were identified with this part.

Q5 This question was based on database theory, choosing data types in Part (a) did not prove enormously challenging; however, some did struggle with the final response.

(b) **(i)** This part of the question on validation discriminated between candidates; many candidates struggled to answer the question. This appeared to be an area where candidates lacked knowledge.

(ii) The majority of candidates were able to identify three appropriate validation rules; however, some seemed to struggle with this part.

(c) This question was answered poorly and suggested a lack of knowledge around this area.

Q6 This question was designed to test the candidates' knowledge of spreadsheets.

(a) This part was well answered.

(b) This was also well answered.

(c) & (d) These two parts were well answered; however, there are still some candidates who use incorrect symbols when writing their answers. Candidates should understand the significance of symbols in spreadsheet formulae.

(e) There were no difficulties with this part.

(f) This question was poorly answered, and some candidates were unable to provide a description of VLOOKUP. This question perhaps needed the VLOOKUP formula to be included, and then the candidate asked to explain the formula.

Q7 Question 7, Parts (a) and Part (b) were well answered, and Part (c) was answered correctly by most candidates.

Q8 **(a)** In Part (a), there were still many candidates who could not expand the letters IoT.

(b) This part was generally well answered. A minority of candidates did not score well in this question.

(c) Candidates generally only scored one mark on this part as they tended to say the same thing and did not give two distinct answers.

(d) This part was well answered.

Q9 This question tested candidates' knowledge about various types of legislation.

(a) This part was poorly answered as candidates did not know the legislation title precisely; therefore, many did not receive a mark.

(b) This part was very well answered.

Q10 This question focused on health concerns related to working on a computer.

Part (a) and Part (b) were poorly answered, some candidates may not have read the question parts correctly and gave the same response for Part (a) and Part (b). Some answers were not specific to the context described in the question and therefore did not gain credit.

Q11 This question was based on cybercrime.

(a) Was well answered.

(b) **(i)** This part was generally well attempted; however, some candidates only mentioned malicious software without any clarification to obtain a second mark.

(ii) This part was well answered.

Q12 The quality of written communication is assessed in this question.

This question is designed to enable the examiner to assess the candidate's ability to write coherently and to discuss such topics. This question required the candidate to describe two disadvantages of online banking to the customer.

Many candidates were able to describe sensible disadvantages. Most candidates were able to write about half a page, with some more able candidates writing more. Very few candidates omitted this question, and it was pleasing to see how it was attempted.

Candidates were able to score into level 2, and many obtained level 3 marks.

Q13 This question was based on definitions, and the candidate had to select the appropriate term from a list of words supplied.

Generally, the question performed well, and many candidates achieved full marks. However, many candidates mixed up the first few definitions. The final five definitions were well answered.

Assessment Unit 2 Digital Authoring Concepts

Unit 1 Overview

This paper contained nine, multi-part, questions. Marks available in various questions parts ranged from one mark to a maximum of eight marks in a question designed to assess candidates' ability to evaluate some of the key interactive features provided in named interactive multimedia applications.

Overall, the paper was found to be accessible and candidates provided a fair and balanced assessment of a broad range of knowledge. Candidates responded positively to most questions posed, with many demonstrating a sound understanding of the key concepts assessed. The structure of the paper provided candidates with a mix of question styles, such as "expand the acronyms," "fill in the blanks," and "match the terms,". This catered effectively to students and ensured the majority of candidates could gain marks across the paper and engage with the content confidently. Extended writing questions required the accurate use of subject-specific terminology to achieve full marks, offering candidates the opportunity to showcase their depth of understanding. Examiners reported however that the level of language used across the paper varied but that a significant number of candidate responses lacked subject specific terminology, and, in some instances, this prevented candidates from accessing higher level marks in some questions. Overall, the examining team was pleased with overall performance and in the main candidate responses indicated that the paper was accessible to all. Most candidates obtained marks in all questions presented across the paper.

It is important, at this point, to remind candidates that when answering questions relating to stimulus material presented in a question, they should make clear reference to the stimuli in their answers, especially if they are asked to do so. For example, in Question 4, where candidates are asked to provide answers to questions associated with labelled database tables and fields candidate responses should relate to the table names and field names provided, otherwise marks cannot be awarded. This is especially important in questions (such as Question 4) where candidates are being asked to identify tables, field headings or criteria used to create a specific query; not doing so will mean the element being developed will not operate correctly and as a result marks will not be allocated.

There appears to be no evidence to suggest that candidates had insufficient time to complete the paper.

Q1 The multiple-choice questions at the beginning of the paper were well answered by many candidates. Some candidates were however less familiar with the key terms assessed in Parts (e) to (g) which focused on key terms associated with database applications.

Q2 The focus on this question was on the tools and techniques associated with designing multimedia and database solutions.

(a) It was refreshing to see candidates show a clear understanding of storyboards used in application design although some did seem to confuse storyboards with navigation design tools with some selecting the second statement as being one which was TRUE when talking about storyboards.

It should be noted that in questions where candidates are asked to place a Tick (√) beside a correct statement, in instances where candidates place more than the required number of ticks, only the first n, where n=the required number, will be accepted.

(b) While many candidates were able to correctly define the term 'target audience' some responses lacked necessary detail in the definition. It is important to reference the idea of a target audience being a 'group of people' as opposed to a person and where a scenario is not provided, marks will not be awarded for describing a potential target audience in this question type.

(c) This question was well answered showing candidates are aware of and able to identify the purpose of each of the design elements listed in the question.

(d) It was clear that not all candidates were familiar with the concept of a data dictionary and its contents. Some candidates referenced designed elements included in storyboards and wireframe diagrams as opposed to data dictionaries.

Q3 This question examined candidate understanding application interfaces.

(a) A significant number of candidates were not able to successfully expand the acronym provided in this question part. It would be expected that all candidates could provide an expansion of GUI at GCSE level.

(b) Only a small number of candidates were able to obtain full marks in this 8-mark question where descriptions of each of the key parts of a WIMP interface were required. In previous examination seasons candidates have shown a clear understanding of the key features of GUI interfaces. Candidates in the main found it difficult to apply any degree of detail in their responses when asked to describe each of the four features listed; except for Pointer which was best described out of all of the four features named.

(c) **(i) & (ii)** Responses provided by candidates tended to focus on the use of voice input in relation to natural language interfaces, despite the fact the question included a prompt referencing the fact that natural language interfaces could also accept typed input. It is important that we broaden candidates' understanding of this concept to ensure they understand the broad range of applications which use natural language input.

Where candidates were awarded full marks in relation to Part (i), the advantages of natural language interfaces, their answers included clear statements which qualified the advantage included in their response. Statements such as 'easier for those with limited mobility' needed more elaboration and should have been followed up with a qualification such as 'as they could speak commands…', while statements such as quicker/easier were not accepted (as has always been the case).

In answering Part (ii), most candidates were able to correctly identify 2 clear disadvantages of using natural language interfaces with the majority of responses focusing on background noise/interference when entering commands and issues around training the device or problems understanding accents. This question part was well answered by most candidates.

(d) Almost all candidates were able to correctly match the identified activities with the most appropriate interface for the application showing there is a clear understanding of where each interface type is most appropriately used when developing applications.

Q4 Questions such as this one, which draw upon candidates, controlled assessment, experiences are generally well answered, and this year was no exception.

(a) Most candidates were able to correctly identify at least two out of the three relationship types associated with relational databases. It should be noted that the only acceptable answers are 1:1, 1:many or many:many.

(b) Many candidates were correctly able to correctly expand the acronym ERD.

(c) It was pleasing to see most candidates were able to correctly identify the primary and foreign keys in the table provided, showing a clear understanding of how relationships are forged between tables in relational database applications.

(d) **(i)** The majority of candidates were able to correctly identify the tables used to create the table shown in the question content when responding to Part (i) of this question.

(ii) When responding to Part (ii) however, it was clear that not all candidates could successfully transfer over the skills developed from controlled assessment query production to successfully design a query which would return the information required by Pampered Pets. Some candidates included customer or pet names in the criteria row which rendered their query incorrect as it would only return 1 result if put into operation in a database application.

A variety of approaches were acceptable as correct responses to their question part. The range of marks awarded was also varied as candidates completed this discriminatory question.

(iii) As with Part (d)(ii) some candidates were unsuccessful in their completion of the design task highlighted in the question stem. Some lost marks by including incorrect criteria e.g. 'cats' as opposed to 'cat', some included 'cat' in one row only with 'Lisburn' and 'Belfast' correctly included in the design grid (meaning only 3 out of the maximum 4 possible marks could be awarded).

A small number of candidates correctly used a logical 'OR' in their criteria in the 'Town' column.

(iv) There is a varied understanding amongst candidates on the use of calculated fields in queries. This is a key part of database design, but it is a difficult concept for some candidates as evidenced by the range of marks awarded to candidates who completed this question part shown. Some candidates completed the question using data items as opposed to field headings and as a result full marks could not be awarded.

(e) It was unfortunate to see a significant number of candidates unable to correctly expand the acronym SQL.

(f) **(i)** Most candidates were able to successfully complete Part (i) of this question where they were required to complete SQL statements to select specific data from identified tables in the database shown.

(ii) Some candidates struggled with this second SQL based question. Only a small number of candidates were awarded the maximum of 3 marks when completing this question. In some instances, candidates selected the incorrect table for inclusion in the FROM statement which in other cases candidates had the correct formula structure but confused > and < meaning the third mark could not be awarded.

Q5 This question which referenced the terms data redundancy and data integrity in relational database applications was the only QWC question in the paper.

Marks awarded varied as is expected with QWC questions. To reach the highest mark band candidates needed to give a detailed definition of both terms with clear examples of each. The majority of candidates had difficulty doing this. Some candidates were able to give clear definitions of each term but did not draw associated examples from the visual prompt provided in the question content. Others attempted to draw examples from the content provided but did not give clear descriptions of the terms. Only a small number of candidates were able to give detailed descriptions and examples of both. Some candidates were able to correctly show the link between data redundancy and poor data integrity using examples shown in the question content. This is to be commended as both can be difficult concepts to verbalise and illustrate.

Q6 This question which referenced multimedia applications from the specification content.

(a) **(i)** Candidates have in the past been asked to define the term multimedia and this question has always been well answered. This year however, there seemed to be only a small number of candidates who had a clear understanding of the idea that multimedia can be defined as the use of 'more than one media', 'such as text, images, video…etc'.

(ii) Similarly, there seems to have been a more limited understanding this year of the term interactive and its definition as shown in previous mark schemes as a smaller proportion of candidates than expected were able to provide a detailed definition of this term.

(b) Candidates were asked in this question to provide a single advantage and disadvantage of each of the interactive features linked to E-commerce and social media applications.

While many candidates were able to provide responses to these question parts, responses were varied in terms of technical content and as a result, marks were also varied. Most candidates were awarded some marks in this question part, however.

In response to online payments, some candidates focused on the digital money transfer (as per shopping in a self-serve terminal) as opposed to being able to pay from anywhere in the world, this was incorrect in the context provided.

In responding to the question part relating to the use of the 360^0 image rotation, some focused on the problems associated with developing the image or storing the image online as opposed to the experience of using the image in the app and as a result their marks were not awarded as the focus of the question was on the experience of using the app.

The question parts relating to tracking, check-in facilities with social media applications were well answered by the majority of candidates, showing candidates are able to draw on real-life experiences to answer some questions (which is exactly in the spirit of this qualification). Only a small number were able to identify appropriate examples of using social media to advertise products/events however, suggesting this is an area where additional focus is needed in content delivery.

(c) In answering this question some candidates lost focus on the 'online interactive' element of online gaming which was being addressed in the question stem and instead focused on general disadvantages of being online. It is important that candidates read each question stem carefully to ensure their responses focus is on the correct concept being assessed.

Q7 This question focused on key specification elements associated with website development.

(a) This question was well answered by the majority of candidates but there is a common misconception that hypertext is code used to create web pages, showing that some candidates confused the term hypertext with HTML.

(b) **(i)** Most candidates were correctly able to identify the need to use folders to help ensure assets can easily be located. It is important that we express to candidates that while folders can help us keep content organized, the reason we do this is to ensure they can be easily located. Questions focusing on why folders are used should prompt candidates to refer to the advantage, i.e. makes them easy to locate, as opposed to the fact that it 'keeps the assets organised'.

(ii) The vast majority of candidates were aware of the need to give files appropriate names but responses such as 'rename them' were deemed to be lacking in detail.

(c) It was pleasing to see the majority of candidates use examples from their controlled assessment tasks to help them successfully complete this question part.

(d) The marking team were also pleased to note that the majority of candidates are now clear on the key features of the four coding constructs identified in the Unit 2 specification.

Q8 This question focused on HTML.

(a) The majority of candidates were able to correctly identify the HTML statement which could successfully display the required image on a web page.

(b) A significant number of candidates still identify <h6> as being the heading style which generates a heading of the largest text size.

(c) In the main, the majority of candidates were able to successfully identify the HTML code which would create the list shown in the question stem.

(d) Candidates are less familiar with the HTML structures associated with the creation of tables with a significant number unable to select the HTML construct used to create the example table shown.

(e) This question was well answered with most candidates being able to identify the graphic as being of bitmap format. It is important however to always remind candidates to check the mark allocation for each question as this is an indicator of the number of correct statements needed to achieve full marks. Some candidates did not take note of the mark allocation in this question and as a result did not make a second correct statement about the graphic file shown in the question.

(f) A large proportion of candidates were unable to select the correct html statement which would correctly add a hyperlink to a web page.

(g) This question which asked candidates to define the term event programming was not well answered in the main. Candidates at GCSE level are expected to be able to clearly define key terms such as this one from the specification.

(h) A large proportion of candidates were unable to expand the acronym CSS. Again, this is a key term which candidates are expected to know and understand at GCSE level.

Q9 The final question of the paper was focused on testing and evaluation.

(a) The term iterative testing is mentioned in the specification but only a small number of candidates were able to give a definition worthy of the 2 marks allocation to this question. Others who answered the question were able to give a broad general description of the term.

(b) Candidates are very familiar with the various testing types covered in the specification, but a significant number of candidates confused Black Box with System Testing.

(c) This question on evaluating multimedia in an application was well answered by the majority of candidates and provided a positive ending to the paper for many.

Principal Moderator's Report

Assessment Unit 3 Digital Authoring Practice

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks via digital submission. Centres **must** ensure candidates have been provided with the correct scenario for Units 3. This changes on an annual basis, released in early June. This information is clearly provided on the front cover of the controlled assessment task. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised. Centres are encouraged however to demonstrate good practice through legacy tasks and provide opportunities for candidates to complete practice tasks.

Administration

The following procedures **must** be adhered to:

- All candidates of the requested sample, including the **top** and **complete bottom candidate** of the sample must be included.
- There is no requirement to copy and paste the **mark band information** into the eCandidate Record Sheet. Brief, factual comments regarding how marking was allocated is sufficient. Only **essential files** should be submitted for moderation, preferably as one **Portable Document Format**. Duplicate word processing files should not be submitted.

Accuracy of Marking

The accuracy of assessment judgements from centres was generally high and demonstrated a sound understanding of the controlled assessment requirements. A small number of centres marked leniently and marking was adjusted. In some cases, marking was slightly harsh at the lower end of the mark range. Centres should use the 'best fit' mark band approach. The use of templates is prohibited. Candidates are encouraged to produce individual solutions. This is a controlled assessment regulation. Assessment of testing and evaluative sections was marked leniently for some centres.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshots or step by step reports for controlled assessment tasks. Thank you to all centres for their efforts this series.

Task 1 Designing the multimedia and database solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- The majority of storyboards were computer generated. Rather than the focus on the volume of storyboarding, candidates should focus on the quality and detail of storyboards, for example, the design of the website could include the design of the masterpage. Database design may include a more text/tabular form/query/report design, to reduce the duplication of work.
- One example of each query, form, report planning is a requirement. More detailed planning on a masterpage would reduce workload, yet still a produce a high quality multimedia solution.
- Storyboards must show the sources of information and file types to be used. Some candidates did not provide the level of detail required - image sources, feedback and prototyping missing from more candidates, some plans were missing sufficient detail for third party implementation.
- The design of professional reports is to be commended, with full titles and the company logo applied appropriately.
- Some candidates failed to plan complex queries (two or more criteria) or appropriate reports (grouping, sorting and calculations).
- Some candidates did not attempt prototyping and the use of feedback in refining the solution, this section was marked leniently. 'Marking up' of prototype amendments and feedback could be applied to storyboards in a different colour.
- Some centres produced excessive detail in their design sections e.g. full prototype designs, followed by full storyboards.
- Centres should avoid creating duplicate versions of storyboards to demonstrate prototyping. Centres should encourage candidates to create a prototype and then include annotated screenshots of the final web pages, highlighting amendments from the original design.

Task 2 Building the multimedia and database solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. The following points should be noted:

- If online web editors are used, e.g. Google Sites, the website URL must be presented for moderation with no additional credentials required to access the site, including access to a C2k account. When using Google Sites with a C2K account it is important to ensure that all content is set to shared access including the pages, the form and all other elements stored Google Drive.
- Graphics, animation and videos produced for the candidate websites were generally of a high standard.
- Overall the standard of web authoring was high. It is positive to note many candidates using masterpages and cascading style sheets.
- Scripted elements included: form validation, slideshow, animation, login, CSS, colour (hex).
- Accessibility elements included: browser settings, plugins (downloads), speech, magnification, alt tags.
- Database solutions incorporated appropriate relationships, user friendly and intuitive switchboards, in addition to appropriate validation and input masks.
- Database solutions were implemented from the design document, however the depth of complex queries and reports (grouping, sorting and calculations) were lacking, particularly grouping and calculations. Some centres included an excessive number of basic queries and reports.

Task 3 Testing the database solution

- This section was marked leniently by some centres.
- Testing must include a range of valid, invalid and extreme data.
- Some candidates designed excessive tests and included screenshots for all testing cases, this could be further refined.
- Corrective action was not completed by a number of centres.
- ***No screenshots are required for testing; screenshots are only required to demonstrate corrective action.***

Task 4 Evaluating the multimedia and database solution

- This section was marked leniently by some centres.
- Many candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.

All documentation should be submitted as one PDF.

Chief Examiner's Report

Assessment Unit 4 Digital Development Concepts

Unit Overview

This unit accounts for 40% of the GCSE Digital Technology (Programming) grade. The total mark for the exam was 120 marks and the time allowed was 90 minutes. The paper contained ten compulsory questions. A range of question types were used in the examination. These included multiple choice, fill in the blanks, closed response, tick box, matching terms and open response questions. Marks allocated to parts of questions ranged from 1 to 9. In Question 7, the candidates' quality of written communication was assessed, this question was worth 6 marks and was marked within levels; there were three levels describing the features of different types of responses from candidates. These extended writing questions require candidates to use the correct digital technology terminology. Candidates making use of generic terms will be unable to score in the top band.

Candidates performed well in questions related to computational thinking. A significant majority of candidates were able to correctly predict output from algorithms provided, identify problems causing logic errors and create accurate solutions to solve problems posed within each of the scenarios on the paper. Responses in this area have improved year on year and it is clear that candidates are being well prepared in centres by their teachers.

A majority of candidates demonstrated a solid understanding of binary arithmetic. Many obtaining full marks in the questions related to this topic.

Questions on arrays continue to provide challenges for some candidates. Where candidates were required to create a loop structure to process the array, many did not use the array structure correctly within the loop e.g. sales[j]. In addition, some candidates did not implement the use of a running total correctly within the loop. As was the case last year, The majority of candidates now recognise that an array is 0 indexed. Some candidates still have difficulty with the use of unbounded iteration when processing array structures. Many prefer the use of bounded iteration for example, the for loop. However, candidates should be able to process an array using other loop types.

Candidates performed well in questions about validation, testing and evaluation.

Responses indicated that candidates were well prepared for the paper and had adequate time to complete it. Questions presented allowed for a broad range of candidates to access marks.

Q1 Question one was a five-part multiple choice question. Candidates performed extremely well in Parts (a), (b) and (c). Parts (d) and (e) provided slightly more challenge. Part (d) required candidates to select the correct table related to Boolean logic. Whilst a significant number of candidates selected the correct answer a common incorrect answer was Part (d).

Part(e) required candidates to know about both the insertion and bubble sorts. This part of the question was less well answered indicating that candidates may need a little more practical experience of these types of sorts.

Q2 This question was devoted to computational thinking and required candidates to predict output and identify problems with algorithmic logic.

Part (a) of this question provided challenge for many candidates. Some provided long lists of numbers for which they received no credit and others provided 4 integers. The correct answer required was a list of three integers 15, 18 and 21. Candidates were awarded full marks if they identified the correct 3 integers.

Part (b) was well attempted despite containing a nested for loop. It was pleasing to see candidates gaining full marks in this question.

Part (c) related to the use of a while loop, the condition in the while loop was incorrect. Candidates, for the most part, identified that the loop would continue to run. Less were able to articulate the change required so that the loop would run correctly. Some suggested that the condition should be changed to “while inches>0” but this was not given any credit as the question clearly stated that the loop should end when a zero was entered.

Q3 Overall, this question was well attempted by candidates.

In Part (a) some candidates lost marks because they did not follow the instructions provided in the question. Reasons included, candidates not outputting the cost of the booking or the category. In some cases, candidates did not construct assignment statements correctly or used more than one condition in each of the if-statements, despite having been instructed to use only one in the question. Candidates should be encouraged to read the instructions provided in the question, particularly the bullet points which clearly identify the requirements for each algorithm. These requirements align closely to the mark scheme.

Parts (b) and (c) where well attempted with many candidates achieving full marks in Part (c) (ii) indicating a solid understanding of the features of syntax and logic errors.

Q4 As in previous years, performance in this type of question was very good indicating great preparation of candidates by their teachers. A significant majority of candidates achieved high marks in the calculation parts of this question. A minority of candidates were unable to perform the hexadecimal conversion. The binary addition question was well attempted with many candidates scoring the full 3 marks.

Part (d) of this question was less well attempted. The question required candidates to identify two differences between Unicode and ASCII code. The mark scheme for this part of the question was broad to enable as many responses as possible to be credited.

Some candidates achieved only two marks as they identified a difference between Unicode and ASCII code and then used the reverse of that difference in the second part of their answer. Common mistakes when answering this question included confusing the number of bits used with the number of characters that each could represent. Generally speaking, candidates were more able to articulate answers about ASCII code. Some responses relating to Unicode indicated that candidates had a very superficial understanding of this method of representing characters.

Q5 This question focused on an algorithm with an associated dry run.

In Part (a) of this question candidates were asked to explain how a dry run could help when testing a program. Whilst a significant majority of candidates were able to explain that a dry run could help identify errors in the logic of the program, fewer were able to make a second point about the dry run. Acceptable answers included “each line of code is reviewed” or “checks to ensure the expected output is produced”. At this level candidates are expected to have a full understanding of the function of a dry run and its role in testing solution design.

Part (b) required candidates to complete a trace table related to the algorithm provided. This part of the question was well attempted by most candidates with many achieving the three marks allocated for inserting the value of variable A. Fewer candidates achieved the mark for predicting the output. Candidates should continue to review examples of code or algorithms using trace tables to help ensure program quality. This will improve performance in this type of question.

Part (c) This part of the question was less well answered. Candidates found it difficult to predict the value of variable A when the loop ended. The correct answer was 16 but a significant number of candidates thought it was 8. This may indicate that candidates find evaluating conditions in a while loop more difficult. Therefore, continued practice in the use of unbounded iteration could be useful.

Q6 This question focused on a scenario where significant detail was given to aid candidates in the completion of algorithms. It should be emphasised that where scenarios are presented candidates are advised to read, carefully, all of the information provided to them. In particular, bullet points which detail important aspects of the solution requirements and which will pertain directly to the mark scheme.

(a) This part of the question was worth 9 marks. Candidates were required to complete an algorithm. This was very well attempted with candidates scoring across the range and many scoring the full 9 marks. Some candidates lost marks in the calculation of TotalCost by entering values rather than variable names for the HireCost and NumberOfNights. Another common mistake was omitting the TotalCost variable in the output statement.

(b) **(i) & (ii)** This question related to Boolean variables and was very well attempted by the majority of candidates showing a clear understanding of the nature and purpose of this data type.

(c) Candidates were required to give one advantage of using a constant. This question was worth 2 marks. However, a significant number of candidates achieved only 1 mark. Many gave an explanation of what a constant is rather than providing an advantage of using a constant. Candidates should be encouraged to make use of constants when programming. A simple use of built-in mathematical constants would suffice to demonstrate the advantages of their use. For example, improved readability or easier maintenance.

(d) **(i)** This was a true or false question relating to validation. It was well attempted with a significant number of candidates achieving full marks. A common incorrect answer was answering false to the statement “validation cannot tell if the data is correct”.

(ii) In this part of the question candidates were required to create an algorithm from scratch given a set of instructions. Again, it is important to emphasise that bullet pointed instructions should be carefully reviewed.

Candidates were required to create a validation loop. The question was reasonably well attempted by candidates. Common errors included, making use of a for loop, incorrectly structuring conditions, making use of the wrong boolean operator and not making use of the variable names provided in the question. It should be noted, that where variable names are provided candidates must make use of them.

Q7 The quality of written communication was assessed in this question. Candidates were allocated marks in one of the four levels.

The question required candidates to simply describe the two programming approaches. There were some excellent responses to this question and many candidates scored in the top band.

A significant proportion of candidates scored in the Level 2 band. It appeared that candidates could more readily describe object-oriented programming and refer to its features more easily. Many made mention of objects, classes, properties, inheritance and even encapsulation. The features of procedural programming were, in some cases, less well described. Some candidates only made reference to procedural programming being a set of step-by-step instructions. Generally speaking, there was a better use of technical terminology in this QWC question than in responses presented in previous years.

Candidates should have a full appreciation of the nature of both approaches to programming.

Q8 This question focused on array structures and candidates were again provided with a detailed scenario.

(a) This was well attempted by the majority of candidates. A significant number achieved full marks however, some candidates answered true to the statements "the contents of an array or list structure cannot be changed" and "the index of an array element stores the value of that element". At this level candidates are expected to have a full understanding of the features of an array structure and how to process the contents of an array.

(b) **(i)** Candidates were provided with full instructions regarding the array name and the requirement to use a loop. Many candidates did not use a loop and some tried to set the array contents to 0 using a simple assignment statement.

Candidates were provided with the array name and some did not use it.

(ii) Candidates were asked to complete a portion of the algorithm relating to the processing of the array contents. Again, this proved challenging for a significant number of candidates. Many candidates did not place the correct variable as the index for sales. A surprising number were unable to complete the running total line for the variable concertOn.

(iii) Candidates were instructed to make use of a while loop to process the tickets sold and the money collected. Again, this question made use of a detailed scenario providing candidates with much of the information relating to the mark scheme. A significant number of candidates were unable to construct a while loop with the correct condition and a counter to move through the array. Most were able to output tickets sold and money collected and could determine that the money collected total was achieved by multiplying tickets sold by two. However, the calculations within the loop were not completed accurately in many cases.

Another point to note with regard to this algorithm is that the money collected calculation and the output statements for tickets sold and money collected should reside outside of the while loop. Candidates should have indicated this in some way, either by using indentation or by explicitly ending the loop. Where candidates placed the output statements inside of the while loop no credit was given. Candidates should be encouraged to indicate the beginning and end of a loop structure so that the logic of their solution is clear.

Q9 **(a)** This Closed response question on test plans was very well attempted. Many candidates achieved the full 7 marks and demonstrated a solid understanding of the role and function of a test plan.

(b) Again this part of the question was well attempted by candidates. A significant number of candidates achieved the full 4 marks indicating that they have a full understanding of the difference between black box and white box testing.

Q10 **(a)** This question asked candidates to apply their understanding of user requirements within a specific scenario. Many found it difficult to state how the user requirements document could assist when features are missing, instead choosing to describe what the user requirements document contains. Thus, a significant number of candidates achieved only one mark in this part of the question. This was disappointing as most candidates would have made use of the user requirements for evaluation purposes when developing their coursework.

(b) This was a straightforward question on testing the robustness of a system. Candidates should have referenced the ability of the system to handle high volumes of valid/null/invalid/exceptional data. However, many candidates incorrectly referred to different types of testing or provided only one example of the type of data to be used when testing robustness. Candidates needed to refer to two of valid/null/invalid/exceptional data to achieve the one mark allocated to this point.

Principal Moderator Report

Assessment Unit 5 Digital Development Practice

Controlled Assessment Tasks

All centres successfully submitted controlled assessment tasks via digital submission. Centres **must** ensure candidates have been provided with the correct scenario for Units 5. This changes on an annual basis, released in early June. This information is clearly provided on the front cover of the controlled assessment task. Controlled assessment guidelines must be followed throughout tasks and 'modelling' using live data files is not authorised. Centres are encouraged however to demonstrate good practice through legacy tasks and provide opportunities for candidates to complete practice tasks.

Administration

The following procedures **must** be adhered to:

- **Program user names and passwords must be recorded accurately to the e-candidate record sheet.** If this is not completed, moderation cannot continue.
- All candidates of the requested sample, including the **top** and **complete bottom candidate** of the sample must be included.
- There is no requirement to copy and paste the **mark band information** into the eCandidate Record Sheet. Brief, factual comments regarding how marking was allocated is sufficient.
- Only **essential files** should be submitted for moderation. Original data files that have been converted to **Portable Document Format** should not be included.

Accuracy of Marking

The accuracy of assessment judgements from centres was sound and demonstrated a clear understanding of the controlled assessment requirements. High quality work was produced by centres and application of the mark scheme was highly accurate.

The use of templates is prohibited. Candidates are encouraged to produce individual solutions. Assessment of testing and evaluative sections continue to be an area of concern for some centres, regarding leniency.

The majority of centres correctly applied the mark scheme and the banding approach to the assessment of candidate work. It is imperative that **all** centres complete internal standardisation effectively before submitting marks to CCEA. Some centres did not follow the mark scheme and/or adhere to internal standardisation procedures. It was positive to note the range of techniques used by candidates in the completion of controlled assessment tasks. There is no requirement for candidates to produce screenshot reports for controlled assessment tasks. Thank you to all centres for their efforts this series.

Task 1 Design a solution using appropriate tools

Many examples of good practice have been noted in the completion of Task 1 by candidates. The following points should be noted:

- In some instances, user requirements were extracted from the task, without development or interpretation. Target audience needs must also be fully identified.
- The design component was particularly well implemented with high quality flowcharts and pseudo code included effectively.
- Validation and error trapping techniques were designed appropriately, however use of user feedback to refine the solution was omitted by a small number of centres and marked leniently.

Task 2 Building a solution

Similarly, many examples of good practice and unique approaches to the task have been noted in the completion of Task 2 by candidates. Centres are to be highly commended upon their approach to this task. The following points should be noted:

- **Remember, program usernames and passwords must be noted on the eCandidate record sheet.**
- Marking was highly accurate across the sampled centres and the programming solutions were of a high standard.
- Candidates should be encouraged to develop their own program style and content.
- Solutions of the highest standard gave due consideration to system robustness.
- All solutions must be annotated.

Task 3 Testing the solution

- Test plans were well structured, in tabular format (showing tests, expected outcome, actual outcome and fixes), in some cases test plans were cluttered and could be more professionally presented.
- It is very important that testing incorporates a range of tests (black and white box testing) and includes valid, invalid and extreme test data. Extreme testing of data was lacking in a number of centres.
- Some candidates designed excessive tests and included screenshots for all testing cases.
- No screenshots are required for testing; screenshots are required to demonstrate corrective action. Corrective action was not completed by a number of centres.

Task 4 Evaluating the solution

- Some candidates described their solution, rather than develop evaluative language. A summary of strengths, weaknesses and improvements led to descriptive rather than evaluative comments.
- It is recommended that candidates structure their evaluation logically in terms of the extent to which the user requirements have been met: fully met, partially met or not met.
- Top candidates successfully addressed performance and robustness issues, with refinements/improvements clearly identified. Performance, robustness and refinements were not always given in depth consideration.
- A small number of centres marked performance, robustness and refinements section leniently.

All documentation should be submitted as one PDF.

Contact details

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